Ahmed Hamdy

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Professional Summary

Final-year Software Engineering student at Universiti Teknologi Malaysia (**CGPA: 3.74/4.00, Dean's List**) with 6 months of hands-on **Software Engineering, DevOps, Cloud, and Platform Engineering** experience at **PETRONAS Digital**. Experienced in building **CI/CD pipelines, Infrastructure as Code, Linux automation, containerized systems, API test automation, distributed systems, and .NET backend services** using Azure DevOps, Terraform, Docker, Kubernetes, GitHub Actions, Redis, PostgreSQL, and RabbitMQ. Independently architected and deployed a **9-service event-driven microservices platform** and am migrating the platform to Kubernetes using Helm, ArgoCD, and GitOps. Seeking graduate or junior **Software Engineer, DevOps Engineer, Cloud Engineer, Platform Engineer, or Site Reliability Engineer** roles in Malaysia.

Technical Skills

Programming: Python, TypeScript, JavaScript, C#, Java, Groovy, SQL, Bash, C++

Backend & APIs: ASP.NET Core / .NET, Node.js, Express.js, FastAPI, Flask, REST APIs, GraphQL, Microservices, Event-Driven Architecture

Frontend: React, Angular, TypeScript, HTML, CSS

Cloud & DevOps: Azure DevOps, GitHub Actions, Docker, Kubernetes, Helm, ArgoCD, Terraform, GitOps, CI/CD, Infrastructure as Code (IaC), Linux, Git, YAML, Shell Scripting, DigitalOcean

Cloud Platforms: Microsoft Azure (VMs, Blob Storage, Container Registry, Pipelines), AWS (CloudWatch)

Databases & Caching: PostgreSQL, MongoDB, Redis, Firebase, Hasura GraphQL

Messaging & Streaming: RabbitMQ, Apache Kafka, Zookeeper

Testing & QA: Groovy/Spock, Jest, Unit Testing, Integration Testing, API Testing, Regression Testing, Mocking, Test Automation, Contract Validation

Observability: Serilog, Structured Logging, Azure Blob Logging, AWS CloudWatch, Monitoring, Logging, Error Handling

Infrastructure Fundamentals: Linux Administration, DNS, TCP/IP, TLS/SSL, Networking, Containers, Distributed Systems, Secrets Management

AI / ML: TensorFlow, Keras, MediaPipe, PyTorch, OpenCV, Groq API

Professional Experience

Software Engineering Intern – DevOps Engineering Sep. 2025 – Mar. 2026
PETRONAS Digital *Kuala Lumpur, Malaysia*

- Built **Azure DevOps YAML CI/CD pipelines** to export, transform, validate, and deploy **Hasura GraphQL metadata** across Development and Production environments, reducing manual deployment steps and configuration drift across enterprise services.
- Automated infrastructure and deployment environments using **Terraform, Linux, Bash, Docker, and Azure DevOps**, including reusable configuration for Linux VM and self-hosted agent workloads operating inside restricted private network environments.
- Developed a reusable **Groovy/Spock API regression testing framework** for automated Dev-vs-Prod comparison, nested JSON normalization, schema validation, data filtering, and endpoint verification across multiple microservices; integrated test execution directly into CI pipelines.
- Engineered backend caching and resilience components for **.NET** services using **Redis, HybridCache, dependency injection, and asynchronous service patterns**, supporting distributed and in-memory caching strategies.
- Implemented centralized structured logging using **Serilog** with console, file, Azure Blob Storage, and relational database sinks, including fallback handling to improve application observability and resilience during external service failures.
- Created automated tests for **.NET backend** and **Angular frontend** components using dependency mocking, API validation, and UI event testing while delivering work in 2-week Agile sprints through **Azure Boards**.

Selected Engineering Projects

StyleSense – AI-Powered Microservices Platform Jan. 2025 – May 2026

- *Node.js, TypeScript, FastAPI, PostgreSQL, Docker, RabbitMQ, Redis, Stripe, GitHub Actions, DigitalOcean*
- Architected and deployed a **9-service event-driven microservices platform** running across **14 Docker containers** on DigitalOcean, supporting salon discovery, booking, payments, messaging, notifications, inventory, and AI-powered recommendations.
- Designed distributed communication using a **RabbitMQ topic-exchange event bus** and implemented a **Redis-backed phantom-token authentication layer** shared across backend services, improving service isolation and centralized authentication.
- Implemented a production-oriented **Stripe Connect payment workflow** using manual payment capture, idempotent webhook processing, and booking state transitions to maintain transactional consistency across distributed service boundaries.
- Built a multi-repository **GitHub Actions CI/CD architecture** across 9 repositories where individual service pipelines trigger a central deployment orchestrator, supporting automated cross-service deployments and manual full-stack releases.
- Trained an **Xception CNN** face-shape classification model on approximately 4,000 images using TensorFlow/Keras and MediaPipe Face Mesh, then integrated model inference with the Groq LLM API for personalized hairstyle recommendations.

Kubernetes & GitOps Platform Migration – StyleSense Aug. 2026 – Present

- *Kubernetes, Helm, ArgoCD, GitOps, Sealed Secrets, k3d, k3s*
- Migrating the 9-service Docker Compose architecture to **Kubernetes**, authoring reusable **Helm charts** for application deployments and standardizing Kubernetes configuration across services.
- Implementing **ArgoCD GitOps** continuous delivery, environment-based Helm configuration, and **Sealed Secrets** for encrypted Kubernetes secret management, targeting a production k3s VPS deployment.

Real-Time Distributed Analytics System Jun. 2025 – Aug. 2025

- *Apache Kafka, Zookeeper, Flask, Java, Maven, Docker, REST APIs*
- Engineered a distributed event-streaming system using **Apache Kafka and Zookeeper** processing more than **1,000 metrics per second**, with containerized service deployment and fault-tolerant message processing.
- Developed a **Flask REST API** to expose real-time aggregated data to a JavaScript dashboard and implemented service health checks and automated recovery logic for containerized workloads.

Education

Universiti Teknologi Malaysia (UTM) Expected Oct. 2026

Bachelor of Computer Science (Software Engineering) – CGPA: 3.74/4.00 *Johor Bahru, Malaysia*

- **Dean's List High Honors**
- Relevant Coursework: Data Structures & Algorithms, Software Engineering, Software Quality Assurance, Distributed Systems, Database Systems
- Selected to exhibit **StyleSense** at **DIGITEX ASCEND 2030**, a national technology exhibition.